

Agriculture: Potato Pest Management Guidelines

Preplant

This year-round IPM program covers the major pests of both commercial and seed potato production fields in California.

About Preplant

- [Mitigate pesticide effects\(/agriculture/potato/Pesticide-application-checklist/\)](/agriculture/potato/Pesticide-application-checklist/) on air and water quality.

What should you be doing during this time?

Consider [crop rotation\(/agriculture/potato/Crop-Rotation/\)](/agriculture/potato/Crop-Rotation/) for reducing pathogens, harmful nematodes, and problem weeds.

Manage weeds in the previous crop.

- [Survey for weeds\(/PMG/C607/m607pcweedsurvey.html\)](/PMG/C607/m607pcweedsurvey.html) and keep records (example [weed survey form\(/PMG/C607/potato-weedsurvey.pdf\)](/PMG/C607/potato-weedsurvey.pdf) PDF).
- Also survey adjacent areas and crops for [problem weeds\(/PMG/C607/m607epprobweeds.html\)](/PMG/C607/m607epprobweeds.html), host plants of green peach aphid, potato psyllid, and other insects or diseases, and volunteer potatoes.
- Control volunteer potatoes.

Avoid herbicides that leave carry-over residues which may affect potatoes.

Take soil samples to [assay for nematodes\(/agriculture/potato/Nematodes/#FIELD\)](/agriculture/potato/Nematodes/#FIELD) and soilborne fungal pathogens immediately after harvest of previous crop.

Manage residue from the previous crop.

Manage weeds between crops.

- Irrigate to stimulate weed seed germination after previous crop, before field preparation.
 - [Cultivate\(/agriculture/potato/Integrated-Weed-Management/#BEFORE\)](/agriculture/potato/Integrated-Weed-Management/#BEFORE) or apply [preemergence herbicide\(/agriculture/potato/Integrated-Weed-Management/#BEFORE\)](/agriculture/potato/Integrated-Weed-Management/#BEFORE) as needed to control weeds.
-

Analyze soil for fertility, physical constraints, and pH, which can affect [common scab\(/agriculture/potato/Common-Scab/\)](/agriculture/potato/Common-Scab/). Fertilize according to soil analysis results. Providing adequate fertility can help with [early blight\(/agriculture/potato/Early-Blight/\)](/agriculture/potato/Early-Blight/) control.

Watch for [wireworms\(/agriculture/potato/Wireworms/\)](/agriculture/potato/Wireworms/), especially if there is a history of damage. Manage if needed according to the Pest Management Guidelines.

Consider soil fumigation for nematodes and some soilborne pests.

- [Nematodes\(/agriculture/potato/Nematodes/#MANAGEMENT\)](/agriculture/potato/Nematodes/#MANAGEMENT) and [Corky Ringspot\(/agriculture/potato/Corky-Ringspot/\)](/agriculture/potato/Corky-Ringspot/)

- [Powdery scab\(/agriculture/potato/Powdery-Scab/\)](/agriculture/potato/Powdery-Scab/)
- [Sclerotium stem rot\(/agriculture/potato/Sclerotium-Stem-Rot/\)](/agriculture/potato/Sclerotium-Stem-Rot/)
- [Verticillium wilt\(/agriculture/potato/Verticillium-Wilt/\)](/agriculture/potato/Verticillium-Wilt/)
- Weed control (indirect benefit)

Make sure to check fumigant label for details on the minimum time required between application and planting to prevent crop damage.

Monitor vole activity in adjacent fields and non-crop areas. Implement baiting programs or install exclusionary fencing if needed to keep voles from moving into potato fields.

Order [seed potatoes\(/agriculture/potato/Seed-Certification-and-Seed-Tuber-Handling/\)](/agriculture/potato/Seed-Certification-and-Seed-Tuber-Handling/) of desired cultivar and certification level.

Check seed tubers for [tuberborne diseases\(/PMG/C607/m607pptuberdisease.html\)](/PMG/C607/m607pptuberdisease.html) and frost damage.

- Bacterial diseases ([Bacterial ring rot\(/agriculture/potato/Bacterial-Ring-Rot/\)](/agriculture/potato/Bacterial-Ring-Rot/), [Soft rot\(/agriculture/potato/Bacterial-Soft-Rot-and-Blackleg/\)](/agriculture/potato/Bacterial-Soft-Rot-and-Blackleg/))
- [Black dot\(/agriculture/potato/Black-Dot/\)](/agriculture/potato/Black-Dot/)
- [Common scab\(/agriculture/potato/Common-Scab/\)](/agriculture/potato/Common-Scab/)
- [Fusarium dry rot\(/agriculture/potato/Fusarium-Dry-Rot-and-Seed-Piece-Decay/\)](/agriculture/potato/Fusarium-Dry-Rot-and-Seed-Piece-Decay/)
- [Late blight\(/agriculture/potato/Late-Blight/\)](/agriculture/potato/Late-Blight/)
- [Powdery scab\(/agriculture/potato/Powdery-Scab/\)](/agriculture/potato/Powdery-Scab/)
- [Rhizoctonia stem and stolon canker \(black scurf\)\(/agriculture/potato/Stem-and-Stolon-Canker/\)](/agriculture/potato/Stem-and-Stolon-Canker/)
- [Silver scurf\(/agriculture/potato/Silver-Scurf/\)](/agriculture/potato/Silver-Scurf/)

Follow proper [handling\(/agriculture/potato/Seed-Certification-and-Seed-Tuber-Handling/#HANDLING\)](/agriculture/potato/Seed-Certification-and-Seed-Tuber-Handling/#HANDLING) procedures for seed tubers.

Consider seed treatments to reduce disease problems.

- [Fusarium seed piece decay\(/agriculture/potato/Fusarium-Dry-Rot-and-Seed-Piece-Decay/\)](/agriculture/potato/Fusarium-Dry-Rot-and-Seed-Piece-Decay/)
- [Rhizoctonia stem and stolon canker \(black scurf\)\(/agriculture/potato/Stem-and-Stolon-Canker/\)](/agriculture/potato/Stem-and-Stolon-Canker/)
- [Silver scurf\(/agriculture/potato/Silver-Scurf/\)](/agriculture/potato/Silver-Scurf/)



UC IPM Pest Management Guidelines: Potato
UC ANR Publication 3463

[Authors and Credits \(../authors-and-credits\)](/authors-and-credits)

Text Updated: 08/19